Assignment 3: Data Visualization

ITCS448: Data Warehousing and Business Intelligence

Faculty of Information and Communication Technology, Mahidol University

Brazilian E-Commerce Public Dataset by Olist

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Section: 1

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Introduction:

Data visualization is a key component of Business Intelligence, transforming complex datasets into clear, actionable insights. This report applies these techniques to a large-scale e-commerce dataset to identify patterns in sales, logistics, and customer behavior that support strategic decision-making.

Data Source:

Dataset: Brazilian E-Commerce Public Dataset by Olist

This report analyzes the "Brazilian E-Commerce Public Dataset by Olist," obtained by Kaggle. The dataset contains 100,000 order records from 2016 to 2018 made at multiple marketplaces in Brazil.

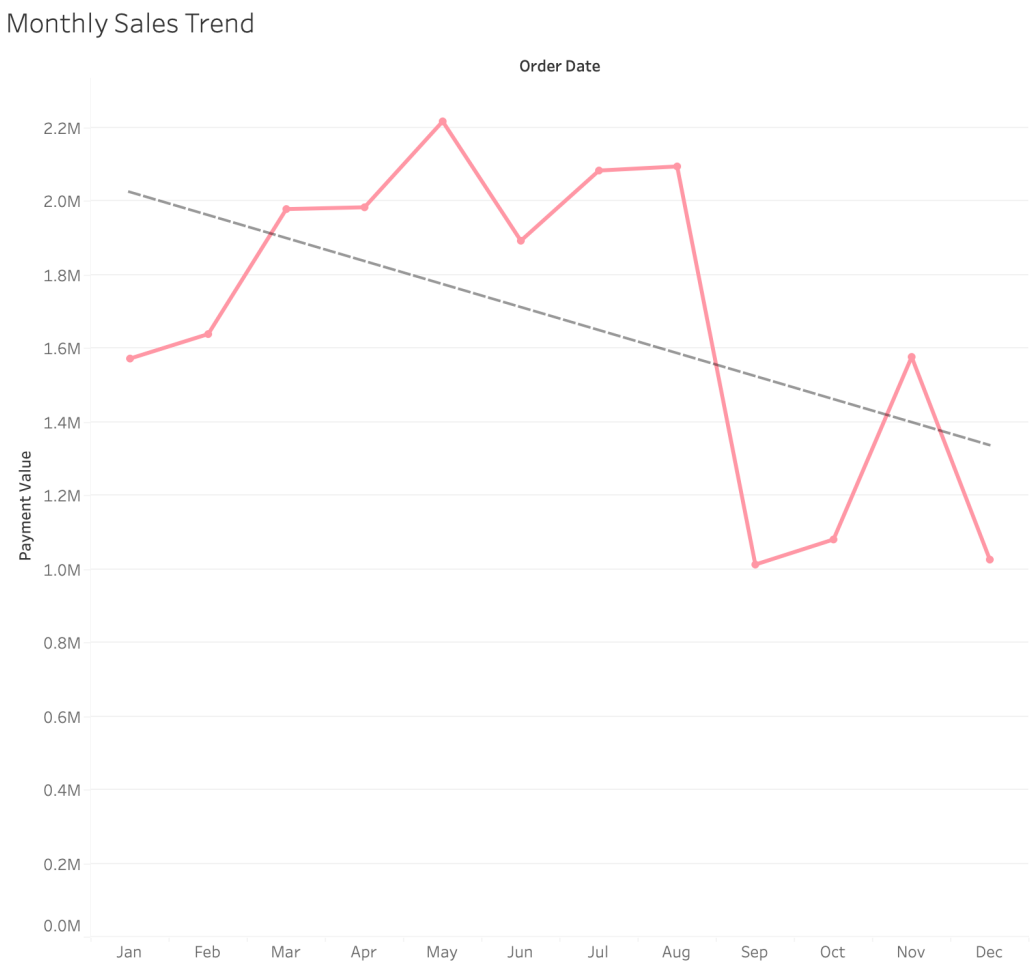
The data is structured into several tables for multidimensional view, including *customer, orders, order_items, products, sellers, payments, reviews, product_category,* and *geolocation*. It represents real-world commercial data, providing insights into order status, price, payment, and freight performance.

Objectives & Business Insights:

The primary goal of this report is to transform raw e-commerce data into actionable business insights. The analysis focuses on the following key areas:

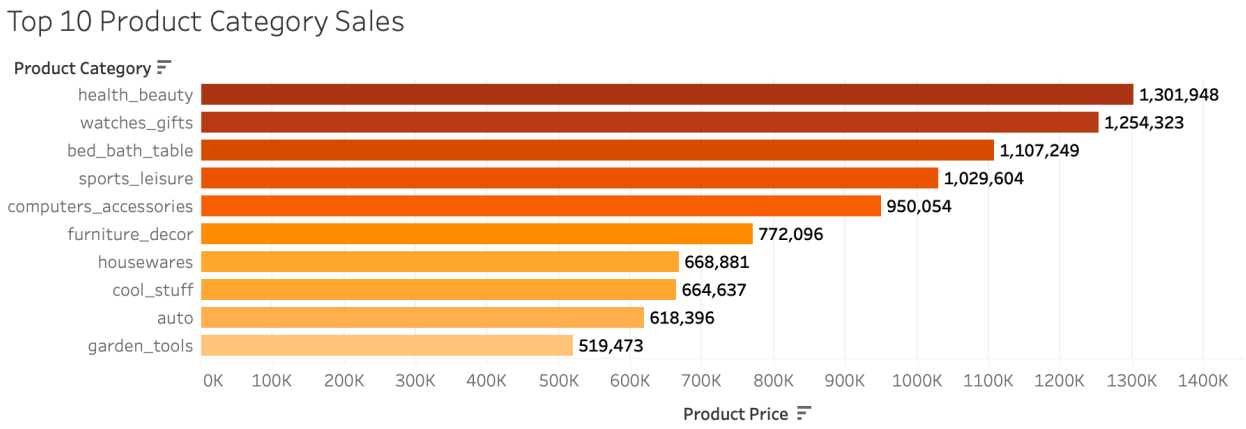
- **Sales Growth & Seasonality:** To identify monthly revenue trends and determine if specific time periods (such as Black Friday) significantly impact sales performance.
- **Product Portfolio Analysis:** To evaluate which product categories are the primary revenue drivers and how they contribute to the overall business volume.
- **Logistics & Operational Efficiency:** To measure delivery performance by comparing actual delivery dates against estimates and analyzing how delays affect customer satisfaction.
- **Customer Experience & Feedback:** To explore the relationship between shipping costs, delivery speed, and review scores to identify areas for service improvement.

Data Analysis:



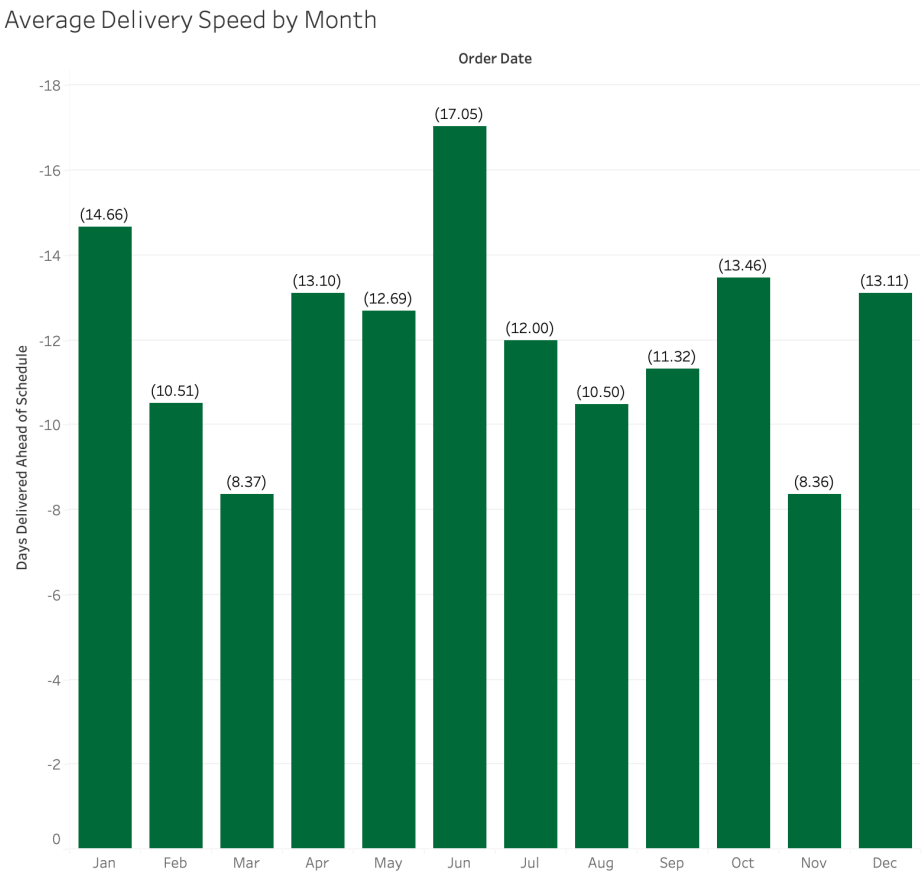
Graph 1: Monthly Sales Trend (Line Chart)

The line chart illustrates the monthly sales performance from 2016 to 2018. There is a clear upward trend in the first half of the period, with a significant revenue peak in May. However, the overall trend line suggests a slight decline toward the end of the observed period. This indicates that while the business experienced strong growth phases, it may face challenges in maintaining consistent sales momentum during off-peak months.



Graph 2: Top 10 Product Categories by Revenue (Bar Chart)

This bar chart identifies the top 10 performing product categories. Health & Beauty emerges as the primary revenue driver, followed closely by Watches & Gifts and Bed, Bath & Table. These top 10 categories represent the core of the company's portfolio. Focusing marketing efforts and inventory management on these high-demand segments is crucial for maximizing future profitability.



Graph 3: Logistics Efficiency - Days Ahead of Schedule (Reversed Bar Chart)

The logistics performance is measured by the average number of days products were delivered before the estimated deadline. Throughout the year, the company consistently achieved ahead-of-schedule deliveries, with an average lead time of 10-17 days earlier than promised. June stands out as the most efficient month. This high level of operational efficiency is a key competitive advantage in enhancing the customer experience.

Graph 4: Impact of Delivery Performance on Customer Satisfaction (Scatter Plot)

The scatter plot examines the correlation between delivery delays and review scores. The trend line shows a negative correlation, meaning that as delivery delays increase, customer satisfaction (review scores) tends to decrease. Most data points are clustered in the 'Early Delivery' and '5-Star' area, confirming that reliable and fast shipping is a significant driver of positive customer feedback and overall satisfaction.

Conclusions:

After examining the Olist e-commerce data, it is evident that the company's strongest asset is its logistics performance. The data consistently shows that most customers receive their orders well ahead of the promised dates, which is directly linked to the high volume of 5-star reviews. However, the business is not without its challenges. There is a noticeable downward trend in sales revenue during the latter half of the period, and the company relies heavily on a few top-performing categories like 'Health & Beauty' to drive its growth. This suggests that while operations are currently efficient, the overall business model needs more stability and a broader variety of successful product segments.

Recommendations:

- **Re-evaluate the Sales Strategy:** Given the recent decline in revenue, it is essential to dig deeper into why sales are slowing down. Whether it's due to seasonal changes or a shift in market demand, creating targeted promotions for off-peak months could help turn this trend around.
- **Expand the Product Portfolio:** Since revenue is currently dominated by only a few categories, Olist should look into ways to boost the performance of under-represented segments. Reducing this dependency will make the business more resilient to market shifts.
- **Prioritize Logistics Excellence:** The high shipping speed is clearly a major reason why customers are satisfied. Olist must ensure that this efficiency remains a top priority even as the business scales, as any delays will immediately lead to lower review scores and customer frustration.
- **Target High-Delay Areas:** By identifying the specific regions or sellers that are causing occasional shipping delays, the management can resolve these bottlenecks before they damage the brand's reputation any further.

References:

- Olist E-Commerce Public Dataset: Data sourced from the Brazilian E-Commerce public dataset by Olist on Kaggle (2016-2018).
- Tableau Software: Data preparation conducted via Tableau Prep Builder and visualization developed using Tableau Desktop.